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Lena-TRNN: Exploring energy flow for time series prediction

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ABSTRACT

We focus on exploring the inherent energy flow for time series prediction in this paper, i.e., we consider the inherent energy of time series data as a sequence measuring properties such as fluctuations, oscillations, and trends. Distinctive with main-stream methods that adopt complex architecture to capture the presentative data relation, this brand-new perspective allows us to better differentiate the underlying distribution of the time-series data. Concretely, we design a novel decoder-free architecture, Latent-energy-aware Transformer Recurrent Neural Network (Lena-TRNN), for multivariate time series forecasting and imputation. The new network tends to assign low energy scores to the samples belonging to the in-distribution dataset, and high energy scores otherwise, which is in accordance with the law of natural change. Predicted samples can be readily obtained by iterating gradient-based optimization along the direction of energy minimization to explicitly learn the inherent energy flow. With the energy optimization modelling, our proposed method exhibits superior performance to many other competitive methods and attains state-of-the-art performance in many benchmark time series forecasting and imputation tasks. The code is available at https://github.com/PengleiGao/Lena-TRNN.

1. Introduction

Interweaving intricate temporal and spatial patterns, time series prediction has been a challenging topic in many real-world applications, such as stock price prediction (Wang et al., 2019), weather forecasting (Zhang et al., 2019), traffic network planning (Chen et al., 2021), and medical diagnosis (Chung et al., 2014). Improving prediction accuracy is important for decision-making and risk management in industrial applications, e.g., long-term forecasting and missing value imputation.

There are two essential questions for time series prediction. 1) How to appropriately represent the temporal variation with sufficient semantic information, which can reflect the inherent properties of time series, such as continuity, periodicity, and trend. 2) How to model the global dependence among different series when processing multivariate time series with high dimensionality. Benefiting from the rapid development and non-linear modeling capacity of deep learning, many works have been proposed to handle the complex temporal variations and spatial correlations for time series prediction. Recurrent Neural Networks (RNNs) (Hochreiter & Schmidhuber, 1997) are designed to model the successive relation between adjacent time points, and Temporal Convolutional Networks (TCNs) (Bai et al., 2018) can capture temporal information by leveraging the 1D convolution kernel along the time dimension. The recent Transformer architectures (Vaswani et al., 2017) with attention mechanisms have been widely applied for time series prediction, which could capture the pair-wise temporal dependencies among time points and spatial correlations among different variables. Examining the current frameworks, it turns out hard for attention mechanisms to find reliable dependencies directly from scattered time points, since the temporal dependencies can be obscured deeply in intricate temporal patterns (Wu et al., 2021). Concurrently, the aforementioned mechanisms are insufficient for representing complex variation patterns and have limitations in making accurate predictions.

Distinctive with the above-mentioned main-stream methods, in this paper, we focus on exploring the inherent energy flow that is considered a sequential measure of properties for time series prediction, such as fluctuations, oscillations, and trends. In particular, we draw inspiration from Energy-Based Models (EBMs), a class of generative models that are constructed based on an energy function, and devise a more elegant framework, named Latent-energy-aware Transformer Recurrent Neural Network (Lena-TRNN), for multivariate time series forecasting and imputation. We illustrate our energy optimization scheme in Fig. 1, where our method aims to find the lowest energy in prediction.

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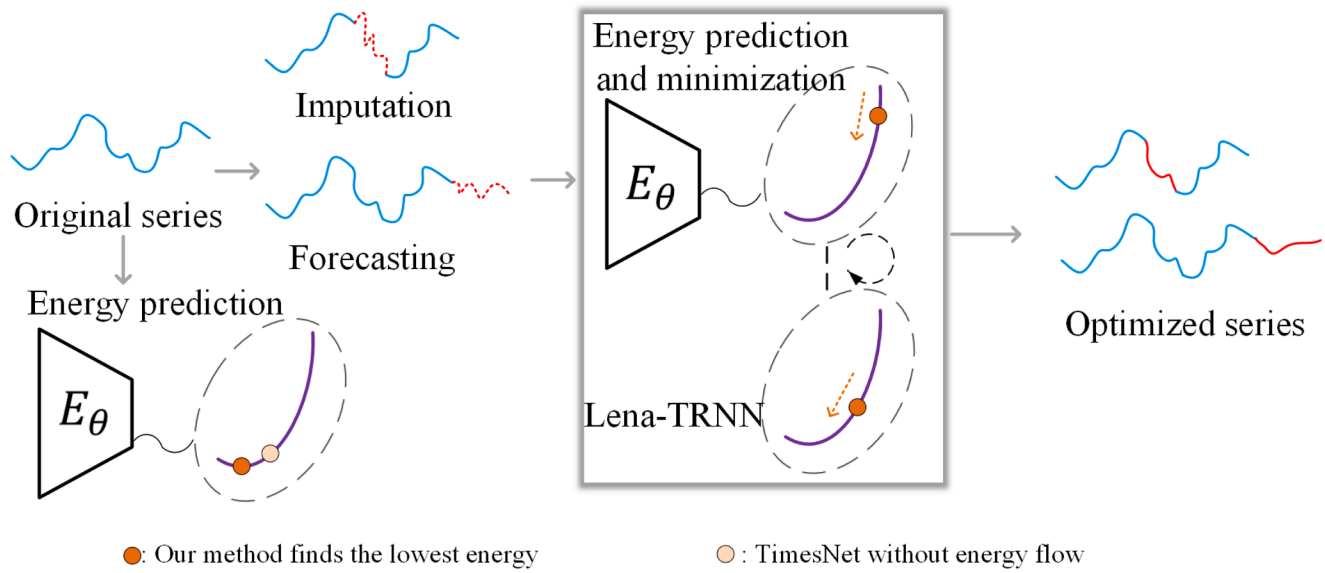
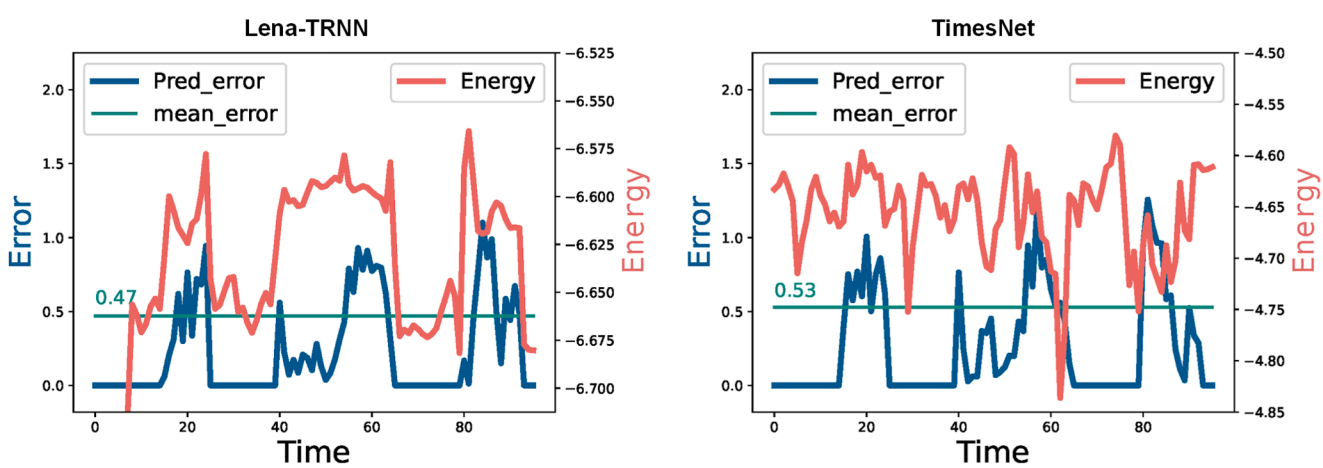


(a) Energy optimization scheme of time series data. The red dashed parts represent the missing data that need to be imputed or predicted, and the red solid line represents the final results.



(b) Visualization of prediction error and energy score. Left: Results of proposed Lena-TRNN; Right: Results of TimesNet.

Fig. 1. (a) The masked time series in imputation or forecasting can be considered corrupted out-of-distribution data. Such inputs can be recovered to the normal series by updating iteratively along the direction of energy minimization. (b) The energy score is illustrated with a negative log operation. The prediction error and energy score clearly correlate in the proposed Lena-TRNN, as shown in the left part. Time steps with larger prediction errors have larger energy scores and vice versa. Such a correlation does not appear in TimesNet (Wu et al., 2023) shown in the right part.

Compared with the previous time series prediction methods, which do not consider energy flow, our method can model the inherent energy variation and learn the trends of energy flow. As shown at the left of Fig. 1, higher energy scores are observed in areas that require missing values imputation having the non-zero error, such as time steps 15–25, whereas lower energy scores are associated with intervals where imputation is not required having zero error. The zero prediction error means that there are no missing values in that interval. Our method demonstrates a consistent relation between predicted error and energy score, i.e., time steps with larger prediction errors have larger energy scores and vice versa. However, such a relation cannot be observed in other methods, such as TimesNet, as shown at the right of Fig. 1. Furthermore, by optimizing energy minimization, our proposed Lena-TRNN effectively discerns the characteristics of time series and has a lower

predicted mean error. The deficiency in recognizing energy variation leads to an inaccurate apprehension of data distribution, consequently resulting in inferior performance. Therefore, discovering the potential energy flow in time series is essential and viable.

Technically, according to the above motivations, we design a recurrent energy flow optimization with the gated recurrent unit (GRU) (Cho et al., 2014) by capturing the temporal involution and local energy score along the time dimension. We apply a linear layer as an energy head to learn the energy score adaptively in the latent space and update the input data iteratively to the target. The recurrent block with time-varying can explicitly learn this kind of implicit energy variation. We further design a Transformer block to capture the global information and energy score of the whole time series data with the self-attention mechanism. The self-attention mechanism can make interactions among different series and obtain global dependence. The proposed framework is a decoder-free architecture and makes predictions with satisfying quality using as few as one gradient updating step. By modeling the inherent energy of input data, we aim to assign low energy scores to in-distribution samples and high energy scores otherwise. Thus, we can perform missing values imputation and time series forecasting with high-energy masked input data. Extensive experiments are conducted on real-world data to verify the efficiency of our method. Our contributions are summarized as follows:

- Motivated by EBMs in image processing on high-dimensional data, we explore the inherent energy variation for time series prediction and explicitly model this energy flow in the latent space. The designed model is optimized along the direction of energy minimization. By differentiating the underlying distribution of the time series data, our method can make accurate predictions for imputation and forecasting tasks.
- We propose a novel framework, Latent-energy-aware Transformer Recurrent Neural Network (Lena-TRNN), to capture the global representation of time series data with a Transformer block and to learn the local energy flow in each time step in a recurrent network. This decoder-free architecture is trained to assign lower energy scores to in-distribution data and higher energy scores to out-of-distribution data, which can be directly used during inference for optimization.
- We conducted extensive experiments on real-world data sets to demonstrate superior performance and illustrate the energy learning during training and inference. Our proposed method achieves state-of-the-art in most mainstream time series prediction tasks.

2. Related works

2.1. Time series prediction

For time series prediction, many classical methods, such as ARIMA (Box & Jenkins, 1968) and VAR (Sims, 1980), have achieved promising results, however, they cannot model nonlinear relationships for multivariate time series and have limitations in their practical applicability. In recent years, deep-learning-based models have been developed for sequential modeling due to their ability to deal with nonlinearities (Huang et al., 2019; Yang et al., 2009) and are widely applied for time series forecasting (Gao et al., 2024a; Lai et al., 2018b; Rangapuram et al., 2018; Yoon et al., 2018) and imputation (Luo et al., 2018; Tashiro et al., 2021). The recent work TimesNet (Wu et al., 2023) achieves state-of-the-art time series forecasting and imputation by transforming the 1D time series into a set of 2D tensors to discover the multi-periodicity. Transformers Vaswani et al. (2017) based on the self-attention mechanism have shown great performance in time series forecasting (Liu et al., 2021; Wu et al., 2021; Zhou et al., 2021, 2022). The long-range dependence of sequential data could be captured with the global modeling capacity. Besides, two works (Zeng et al., 2022; Zhang et al., 2022) use Multi-layer Perceptron (MLP) to model temporal dependencies and achieve promising results compared with the transformer architecture. The authors in Gao et al. (2024b) introduced the Diffusion Model to generate stock factors with Transformer architecture to enhance the training procedures. This method incorporated conditional guidance to capture global joint distribution and model the trend of the stock price. The work (Chen et al., 2024) introduced a conditional diffusion framework incorporating disentangled temporal representations, such as trend and seasonality, as conditional information and employed contrastive learning to improve generalizability. This method is mainly designed for temporal-spatial data which has a strong correlation in spatial connection, such as traffic network sensor data. Yet, none of the existing methods considers the inherent energy flow of time series data and makes predictions based on energy optimization, which could help to better differentiate the underlying distribution of time-series data.

2.2. Energy-based models

Our proposed recurrent framework with energy flow is based on the progress of EBMs (LeCun et al., 2006) that aim to perform probabilistic modeling over data and conditional sampling with an energy function. The energy function is used to measure the compatibility between the input data and the model's predictions. The ideas of EBMs have been widely used in a variety of applications, including image generation (Wang et al., 2023), anomaly detection (Dehaene et al., 2020), and latent space modeling of generative models (Pang et al., 2020). The authors in (Du & Mordatch, 2019) presented techniques to scale MCMC based on EBMs training for robotic hand trajectories in a non-autoregressive fashion. The authors in the work of Kong et al. (2020) proposed a versatile diffusion probabilistic model for conditional and unconditional waveform generation. Although these methods leveraged EBMs to make predictions, they did not consider the temporal development of the input data. The work in Brakel et al. (2013) presented a strategy for training energy-based graphical models for time series imputation. The inference is conducted by either truncated gradient descent or variational mean-field iterations. Two efforts (Rasul et al., 2021a; Yan et al., 2021) used diffusion and score-matching methods for probabilistic forecasting with multivariate time series. They model the data distribution with an autoregressive scheme implicitly learning the diffusion noise, and convert white noise into a sample of the target distribution through Langevin sampling. Different from the above methods our proposed Lena-TRNN is a decoder-free architecture and makes predictions based on the explicit energy flow of input data without extra sampling. To the best of our knowledge, our method is the first to explore the inherent energy flow to guide the capture of temporal variation and global dependence for time series prediction. The energy function in our method can be any function that maps input data to a scalar value. This allows using complex architectures such as neural networks to represent the energy function. Additionally, EBMs can handle continuous and discrete variables without any changes to the model.

3. Method

In this section, we will introduce in detail our proposed energy flow optimization for time series modeling. We first review the backgrounds of time series prediction modeling and energy-based models in Sections 3.1 and 3.2, respectively. We then present the proposed Lena-RNN in Section 3.3.

3.1. Problem statement

In time series prediction, we have time series data as $\boldsymbol{X} = \boldsymbol{X}_1, \ldots, \boldsymbol{X}_t, \ \ldots, \boldsymbol{X}_{\mathcal{T}} \in \mathbb{R}^{\mathcal{T} \times K}$ representing K multivariate time series

sampled for total $\mathcal{T}$ time steps, where $x_t^k \in \mathbb{R}$ is the observed value of the kth variable at time point t.

For time series imputation, we aim to build an appropriate model to predict the missing values of the input data $\boldsymbol{X}_{1:T}$ with length T. Also, we have an observation mask $\boldsymbol{M} = \{m_{1:T,1:K}\} \in \{0,1\}^{T\times K}$, where $m_{t,k} = 0$ if the input data x_t^k is observed, and $m_{t,k} = 1$ if x_t^k is missing. The time interval can be different between two consecutive observed data. The imputed output is obtained with $\hat{\boldsymbol{X}}_{1:T} = f_\theta(\boldsymbol{X}_{1:T}, m_{1:T,1:K}; \boldsymbol{\theta})$, where f_θ is the designed model and $\boldsymbol{\theta}$ are the learnable parameters. As for the time series forecasting problem, the goal is to predict the future L steps $\boldsymbol{X}_{T+1:T+L} \in \mathbb{R}^{L\times K}$ given the historical H steps of observation data $\boldsymbol{X}_{T-H+1:T} \in \mathbb{R}^{H\times K}$. In practice, the data sample is obtained by sliding window from the whole time series according to the history length H and the prediction length L. With the representation model f_θ, we will have $\boldsymbol{X}_{T+1:T+L} = f_\theta(\boldsymbol{X}_{T-H+1:T}; \boldsymbol{\theta})$.

3.2. Energy-based models

Primarily functioning as generative models, EBMs are often designed to represent a target density distribution function. For any given data sample that has a probability distribution $p_D(\boldsymbol{x})$, the training process for EBMs typically involves learning an energy function that can estimate the unnormalized density, referred to as the energy score. In particular, the estimated probability distribution can be represented as:

$$p_\theta(\boldsymbol{x}) = \frac{\exp(-E_\theta(\boldsymbol{x}))}{Z_\theta}, \tag{1}$$

where $\boldsymbol{x} \in \mathbb{R}^D$ and the energy function $E_\theta(\boldsymbol{x}) : \mathbb{R}^D \to \mathbb{R}$ assigning energy score to the sample, with $\boldsymbol{\theta}$ representing the learnable parameters. $Z_\theta = \int_x \exp(-E_\theta(\boldsymbol{x}))$ is known as the partition function. This score is anticipated to be low for in-distribution (positive) samples and high for out-of-distribution (negative) samples. Usually, according to the maximum likelihood (ML), one can approximate a target data distribution $p_D(\boldsymbol{x})$ by minimizing the expected negative log-likelihood function over the data distribution. The objective function can be formulated as:

$$\begin{aligned}\mathcal{L}_{ML} &= \mathbb{E}_{\boldsymbol{x}\sim p_D(\boldsymbol{x})}[-\log p_\theta(\boldsymbol{x})] \\ &= \mathbb{E}_{\boldsymbol{x}\sim p_D(\boldsymbol{x})}[E_\theta(\boldsymbol{x}) + \log Z_\theta].\end{aligned} \tag{2}$$

However, training an EBM through this ML optimization is difficult, since the computation of $\mathcal{L}_{ML}$ involves the intractable Z_θ. The common practice is to represent the gradient of $\mathcal{L}_{ML}$ as follows:

$$\begin{aligned}\nabla_\theta \mathcal{L}_{ML} = &\ \mathbb{E}_{\boldsymbol{x}^+\sim p_D(\boldsymbol{x})}[\nabla_\theta E_\theta(\boldsymbol{x}^+)] \\ &- \mathbb{E}_{\boldsymbol{x}^-\sim p_\theta(\boldsymbol{x})}[\nabla_\theta E_\theta(\boldsymbol{x}^-)].\end{aligned} \tag{3}$$

Training the model $E_\theta(\boldsymbol{x})$ through Eq. (3) can effectively distinguish in-distribution and out-of-distribution samples by decreasing the predicted energy of positive data samples $\boldsymbol{x}^+$ from the true data distribution and increasing the energy of negative samples $\boldsymbol{x}^-$ generated from the model p_θ.

According to this training strategy, we aim to generate in-distribute samples with low energy scores under data distribution. The energy function is parameterized as a deep neural network, allowing for the learning of a continuous energy space from data, which makes it possible to find low-energy samples by navigating the energy landscape. In previous works, gradient-based optimization methods (Du et al., 2021; Du & Mordatch, 2019) transform the negative sample to the positive sample by moving its energy along the direction of energy minimization. This process is iteratively performed to generate a sequence of samples that converges towards low-energy regions in the energy space. The approximated optimum $\tilde{\boldsymbol{x}}^N$ is obtained by performing N gradient steps:

$$\begin{aligned}&\tilde{\boldsymbol{x}}^n = \tilde{\boldsymbol{x}}^{n-1} - \alpha\nabla_x E_\theta(\tilde{\boldsymbol{x}}^{n-1}) + w^n, \\ &w^n \sim \mathcal{N}(0, 2\alpha), \quad n = 1, \dots N,\end{aligned} \tag{4}$$

where α is the step size of the gradient-based optimization and the noise term w^n is usually set to a smaller scale for faster sampling (Du et al., 2021). The initial value $\tilde{\boldsymbol{x}}^0$ is often obtained by sampling from a prior distribution. In our work, the negative samples are data with missing values for the imputation task and pre-predicted series, such as mean values of previous steps, for the forecasting task.

3.3. Proposed Lena-TRNN

With the property of time series data, we utilize a recurrent neural network to represent the temporal features of time series inputs and a Transformer block to capture the global representation of multivariate series. We apply a linear head h_{linear} on the recurrent feature extractor to construct our proposed energy flow optimization. The energy function can be expressed as $E_\theta(\cdot) = h_{linear}(RNN(\boldsymbol{x}_t, \boldsymbol{c}_t))$ with θ representing the parameters of both h_{linear} and $RNN(\cdot)$, where $\boldsymbol{x}_t$ is the input series and $\boldsymbol{c}_t$ is the global representation of the input data. In a typical setting, the linear head h_{linear} contains only hundreds of parameters. In image processing, the typical implicit sampling of EBMs can take dozens or even hundreds of gradient steps to produce an image sample of satisfying quality (Du et al., 2021; Du & Mordatch, 2019), which will introduce an unaffordable cost. In Wang et al. (2023), they abandon the approach of sampling from scratch and instead train the neural network to conduct conditional sampling, with the goal of reconstructing partially damaged data using direct supervision. Inspired by the idea, we consider the input data with missing values or the pre-predicted series as the corrupted data and obtain the desired time series output moving towards the low-energy region rapidly. The proposed framework can allow the prediction of each sample with as few possible gradient optimization steps and alleviate the unstable EBMs training.

Taking examples of time series imputation, given an input time series sample $\boldsymbol{X}_{1:T}$ and an observation mask $\boldsymbol{M}$ representing where the missing values appear. With different mask ratios, we can get the corresponding input data $\tilde{\boldsymbol{X}}_{1:T}$ with missing values, which can be seen as an out-of-distribution sample of p_D. As mentioned before, we use a linear head to model the inherent energy flow and can perform online evaluations by imputing the incomplete data to the original series $\boldsymbol{X}_{1:T}$ in the modeled energy space along the direction of energy minimization. According to the gradient-based optimization (4), our energy flow optimization can be expressed with N-step restoration:

$$\tilde{\boldsymbol{X}}^j = \tilde{\boldsymbol{X}}^{j-1} - \alpha\nabla_X E_\theta(\mathrm{SG}(\tilde{\boldsymbol{X}}^{j-1})), \quad j = 1, \dots, N, \tag{5}$$

with SG denoting the *stop gradient* operation that blocks the gradient propagation across steps, helping accelerate the training speed and convergence. By adding the *stop gradient* operation which is empirically validated and employed in Chen et al. (2021), Du and Mordatch (2019), Wang et al. (2023), we can ensure that each update step independently considers the current imputed series as a fixed point for gradient calculations, avoiding the computational complexity of backpropagating through all previous iterations. In the inter-loop of energy updating, we want to keep the stability of model training without accessing the gradient of $\boldsymbol{x}$ in each step. E_θ can be considered as an internal loss to explicitly learn the energy flow and α is a learnable coefficient. This optimization scheme encourages higher energy values for negative samples and updates them to in-distributed samples along the direction of energy minimization. We use MAE loss for N steps as our final objective to update the whole network's parameters:

$$\mathcal{L} = \frac{1}{N}\sum_{j=1}^{N} \mathrm{MAE}(\tilde{\boldsymbol{X}}^j, \boldsymbol{X}). \tag{6}$$

The final predicted result $\hat{\boldsymbol{X}}$ is obtained by N iterations $\tilde{\boldsymbol{X}}^N$. The overall architecture is shown in Fig. 2. Different from previous works, our method leads to a decoder-free architecture and makes predictions based on the explicit energy flow of input data. Accurate results can be obtained with few gradient optimization steps directly on the input data without a complex sampling procedure.

To represent the temporal variation of time series data, we apply a GRU module capturing the temporal latent features and updating the incomplete data at each time step. We further utilize a Transformer block

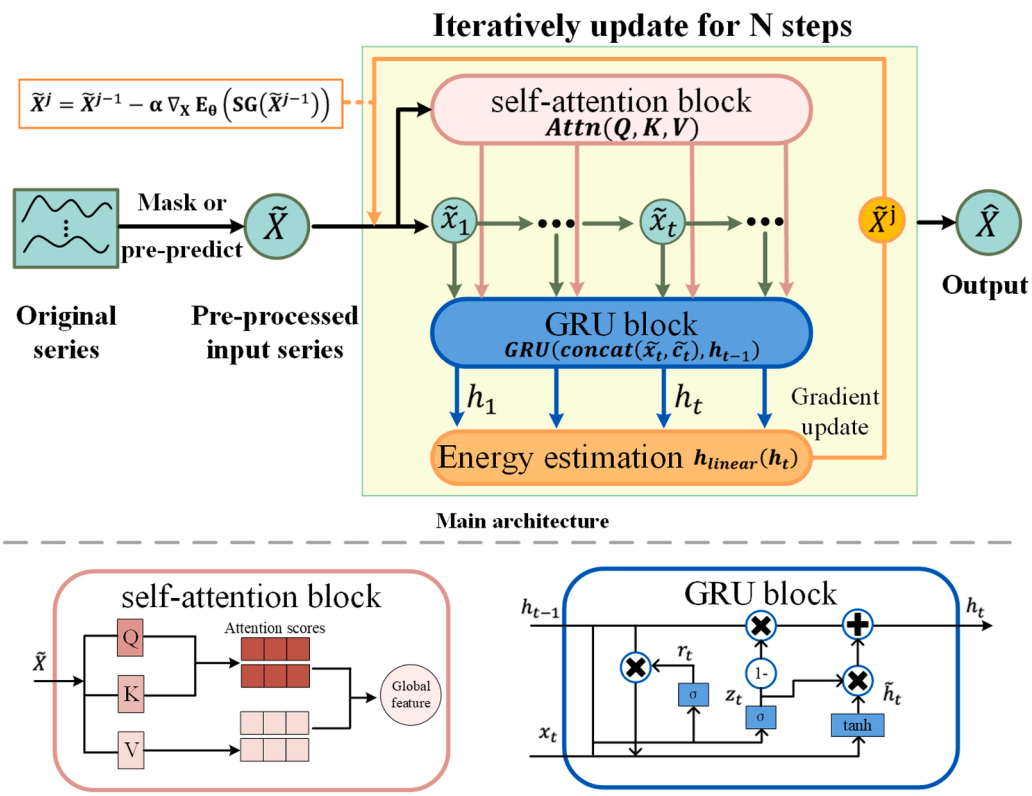


Fig. 2. Top: Our proposed main architecture Lena-TRNN. Bottom: The detailed structures of the designed Transformer block and GRU network. The original series is first masked for imputation or pre-predicted for forecasting to get $\tilde{\boldsymbol{X}}$. Then the Transformer block generates $\tilde{\boldsymbol{C}}$ as the conditional input to the GRU block. Energy estimation approximates the energy score at each time step according to the hidden state $\boldsymbol{h}_t$. Final predicted result $\hat{\boldsymbol{X}} = \tilde{\boldsymbol{X}}^N$ is obtained by iteratively updating based on Eq. (5). The Q, K, and V in the Transformer block represent the Query, Key, and Value representation vector. Latent variables $\boldsymbol{r}_t$ and $\boldsymbol{z}_t$ are the reset gate and updating gate in the GRU cell.

based on the self-attention mechanism to learn the global energy flow of the whole input time series. The global long-range dependence of the multivariate time series can also be captured by the Transformer block ϕ_{att}:

$$\tilde{\boldsymbol{C}}_{1:T} = \phi_{att}(\tilde{\boldsymbol{X}}_{1:T}) = \text{Proj}(\text{Attn}(\tilde{\boldsymbol{X}}_{1:T})), \tag{7}$$

where Attn is the attention mechanism implemented based on Vaswani et al. (2017) and Proj is a linear layer to transform the latent dimension of the attention block to the input dimension. The self-attention block Attn can be formulated as:

$$\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} = \text{MLPs}(\tilde{\boldsymbol{X}}_{1:T}), \tag{8}$$

$$\text{Attn}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Softmax}(\boldsymbol{Q}\boldsymbol{K}^\top/\sqrt{d_k})\boldsymbol{V}, \tag{9}$$

where MLPs represent three multilayer perceptrons. $\tilde{\boldsymbol{C}}_{1:T} = \{\tilde{\boldsymbol{c}}_t\}_{t=1}^T$ is the learned global representation of the whole input series as a conditional input of the GRU block. The hidden state is updated in the recurrent network as:

$$\boldsymbol{h}_t = \text{GRU}(\text{concat}(\tilde{\boldsymbol{x}}_t, \tilde{\boldsymbol{c}}_t), \boldsymbol{h}_{t-1}), t = 1, \dots, T. \tag{10}$$

According to the hidden states at each time step, the energy score is computed as:

$$E_\theta(\cdot) := \text{energy} = h_{linear}(\boldsymbol{h}_t), t = 1, \dots, T, \tag{11}$$

where the linear head h_{linear} is realized with a simple linear layer. The energy flow optimization of the input series can be accomplished by Eq. (5). We summarize the overall training steps of the proposed method in Algorithm 1. Our proposed energy estimation could be seen as a regularization term and estimated with the designed network, enabling direct predictions on the original data through a gradient-based perspective. It guides the model to generate plausible imputations and predictions by explicitly modeling the discrepancy between the predicted and observed time series through an energy-based objective and gradient updating. This encourages the network to learn smooth, consistent representations of the underlying data manifold and to avoid over-fitted solutions. This decoder-free architecture can be adapted as a plug-in to enhance the performance of other methods with suitable modifications.

Algorithm 1 Energy flow optimization for time series.

Given: ϕ_{att}, $GRU(\cdot)$ ▷ Network architecture
Given: N ▷ Number of steps for the gradient update
Input: A pre-process input time series $\tilde{\boldsymbol{X}}$ ▷ masked input series for imputation task or pre-predicted input series with linear layers for forecasting task
Initialization: Step size α, Transformer block ϕ_{att}, GRU block, and Linear Head h_{linear}
repeat
  for Step $j = 1 : N$ **do** ▷ Update gradient in each step
    Stop gradient $\tilde{\boldsymbol{x}}_t^{j-1} = \text{SG}(\tilde{\boldsymbol{x}}_t^{j-1})$ ▷ blocks the gradient propagation at each step j to accelerate the training speed and convergence
    Obtain the conditional input $\tilde{\boldsymbol{C}}$ with $\tilde{\boldsymbol{C}} = \phi_{att}(\tilde{\boldsymbol{X}})$
    Compute the hidden states $\boldsymbol{h}_t$ and corresponding energy score at each time point according to Eqs. (10) and (11)
    Perform gradient update on $\tilde{\boldsymbol{x}}_t^{j-1}$ according to Eq. (5) at each time point to obtain $\tilde{\boldsymbol{x}}_t^j$
  end for
  Get the total MAE loss with Eq. (6) for all steps N and update the parameters of the whole network with gradient descent. Obtain final output $\hat{\boldsymbol{X}} = \tilde{\boldsymbol{X}}^N$.
until Converge

Table 1
Summary of experiment benchmarks.

Benchmarks	# of variables	step of impute/forecast
ETT(4 subsets)	7	96 / 96~720
ECL	321	96 / 96~720
Weather	21	96 / 96~720
Traffic	862	- / 96~720
Exchange	8	- / 96~720
ILI	7	- / 24~60

4. Experiments

In this section, we evaluate the effectiveness of our proposed energy flow optimization on time series imputation and long-term forecasting tasks with extensive experiments on various real-world datasets.

4.1. Datasets

Following the previous works, we select six real-world benchmarks for our experiments. The summary of these benchmarks is shown in Table 1. (1) **ETT** (Electricity Transformer Temperature) (Zhou et al., 2021) dataset is collected from electricity transformers containing load and oil temperature between July 2016 and July 2018, which is a crucial indicator in the electric power long-term deployment. There are four subsets in this dataset, in which {ETTh_1, ETTh_2} represents 1-hour-level and {ETTm_1, ETTm_2} represents 15-minutes-level. (2) **ECL** (Electricity Consuming Load)[1] dataset is collected from the electricity consumption (Kwh) of 321 clients containing hourly data from 2012 to 2014. (3) **Weather**[2] dataset is collected every 10 minutes containing 21 meteorological indicators in 2020, such as air temperature, humidity, etc. (4) **Traffic**[3] dataset is collected from the California Department of Transportation containing hourly data describing the road occupancy rates measured by different sensors on San Francisco Bay area freeways. (5) **Exchange** (Lai et al., 2018b) dataset is collected for the daily exchange rates of eight different countries from 1990 to 2016. (6) **ILI**[4] dataset is collected from the Centers for Disease Control and Prevention of the United States between 2002 and 2021 containing the weekly recorded influenza-like illness (ILI) patients data. We split the dataset with training, validation, and test sets following the previous work (Wu et al., 2023). We select the Mean Squared Error (MSE) and Mean Absolute Error (MAE) as evaluation metrics. For probabilistic forecasting, we select three datasets for comparison including **Solar** (Lai et al., 2018a), **Exchange**, and **Traffic** according to Rasul et al. (2021a).

4.2. Baselines

We select recent state-of-the-art *transformer-based, CNN-based and MLP-based* models for deterministic comparisons on the imputation and long-term forecasting task, including Informer (Zhou et al., 2021), Autoformer (Wu et al., 2021), FEDformer (Zhou et al., 2022), Non-stationary Transformer (Liu et al., 2022), ETSformer (Woo et al., 2022), TimesNet (Wu et al., 2023), LightTS (Zhang et al., 2022), and DLinear (Zeng et al., 2022). We also compare the probabilistic forecasting task with two diffusion-based models TimeGrad (Rasul et al., 2021a) and ScoreGrad (Yan et al., 2021), and one transformer-based model Transformer-MAF (Rasul et al., 2021b).

[1] https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014
[2] https://www.bgc-jena.mpg.de/wetter
[3] http://pems.dot.ca.gov
[4] https://gis.cdc.gov/grasp/fluview/fluportaldashboard.{html}

4.3. Implementation setting

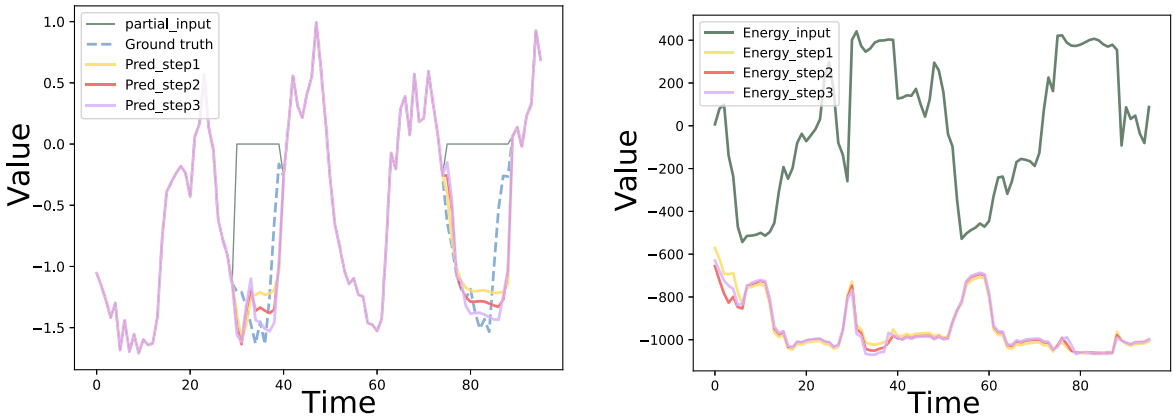
Following the settings of imputation and long-term forecasting in Wu et al. (2023), we select the datasets from five real-world applications as benchmarks, including ECL, ETT, and Weather for imputation, and ECL, ETT, Weather, Traffic, Exchange, and ILI for long-term forecasting. All the samples are obtained by sliding windows. All experiments are repeated three times and we report the average results for all tables. All the experiments are implemented with PyTorch (Paszke et al., 2019) and conducted on a single NVIDIA RTX3090. Our method is trained by using L1 loss with ADAM (Kingma & Ba, 2015) with an initial learning rate of 10^{-2} for ILI, 10^{-3} for ECL, and 10^{-4} for other datasets. The training process is early stopped within 3 epochs. Results of baselines are copied from Wu et al. (2023). The number of self-attention layers is set as 4 and the model dimension is 512 for Electricity (ECL) and 256 for others in the imputation task. The coefficient of energy score is selected in {10,15,20,30}. For the long-term forecasting task, the number of self-attention layers is selected in {2,4}. The model dimension is 512 for Electricity (ECL), 256 for Traffic, and selected in {32,64} for ILI, {128,256} for ETT, Weather, and Exchange. The coefficient of energy score is selected in {10,15,20}. The batch size is set within {32,64,128} for maximizing utilization.

4.4. Main results

Imputation. We randomly mask the time points in the ratio of {12.5 %, 25 %, 37.5 %, 50 %} followed Wu et al. (2023) to compare the model performance under different proportions of missing data. We list the average results of the four different mask ratios in Table 2. Full results of each mask ratio are provided in the Appendix. Our proposed energy-aware method achieves state-of-the-art results for most datasets in this challenging task that has complex temporal patterns in the irregular and partially observed time series. In particular, there are significant improvements of 7.4 %/5.6 % and 6.1 %/6.8 % (MSE/MAE) on the ETTm1 and ETTh2 datasets for the average results compared to the second-best method, respectively. Our method achieves 50 best performance and 10 second-best performance among all the subset results for all datasets. Detailed can be seen in Table A.1. Additionally, our results demonstrate a comparable performance with the TimesNet on the high-dimensional ECL dataset. The learned temporal correlations, derived from the energy-aware model, are found to be competitive with the multi-periodicity based on 2D tensor learning while offering a concise way of learning energy variation. Modeling the joint distribution of the observed and missing data with energy flow enables the capture of effective temporal dependencies, thereby benefiting the prediction of the in-distribution imputed series.

Long-term forecasting. The input length is set as 36 for ILI and 96 for other datasets, and the prediction length is set as {24, 36, 48, 60} for ILI and {96, 192, 336, 720} for other datasets following Wu et al. (2023, 2021). We list the average results of the four different prediction lengths in Table 3. The quantitative results show that our method achieves state-of-the-art results for most datasets. Especially, there are significant improvements of 11.9 %/10.8 % and 8.2 %/6.8 % (MSE/MAE) on ECL and ETTh2 datasets compared to the second-best method, respectively. Our method achieves 50 best performance and 14 second-best performance among all the subset results for all datasets. Detailed can be seen in Table A.2 The results show that our method performs better with high dimensionality on ECL with 321 variables. The energy-aware scheme helps to capture the correlation and inherent energy among variables and optimize each series at each time step along the direction of energy minimization. The results of each prediction length are provided in the Appendix for a more intuitive understanding.

Probabilistic forecasting. We select three datasets that are commonly used for probabilistic time series forecasting in this section. The predicted step is the same as previous studies (Rasul et al., 2021a) and we predict all features at future time steps with historical data. For



(a) Imputation results of different steps. (b) Latent energy score of different steps.

Fig. 3. Visualization of the imputed results and corresponding latent energy score of different updating steps on the ECL dataset for imputation task with 25 % mask ratio.

Table 2
For the imputation task, the data are randomly masked with ratios of {12.5 %, 25 %, 37.5 %, 50 %} in length-96 time series. The experiments were repeated three times and we reported the mean values. The results are averaged from 4 different mask ratios. Full results are provided in Appendix A.1.

Model	Lena-TRNN		TimesNet		Dlinear		ETSformer		FEDformer		Stationary		Autoformer		Informer		LightTS	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	**0.025**	**0.101**	0.027	0.107	0.093	0.206	0.120	0.253	0.062	0.177	0.036	0.126	0.051	0.150	0.071	0.188	0.104	0.218
ETTm2	**0.021**	**0.082**	0.022	0.088	0.096	0.208	0.208	0.327	0.101	0.215	0.026	0.099	0.029	0.105	0.156	0.292	0.046	0.151
ETTh1	**0.074**	**0.182**	0.078	0.187	0.201	0.306	0.202	0.329	0.117	0.246	0.094	0.201	0.103	0.214	0.161	0.279	0.284	0.373
ETTh2	**0.046**	**0.136**	0.049	0.146	0.142	0.259	0.367	0.436	0.163	0.279	0.053	0.152	0.055	0.156	0.337	0.452	0.119	0.250
ECL	0.094	0.211	**0.092**	**0.210**	0.132	0.260	0.214	0.339	0.130	0.259	0.100	0.218	0.101	0.225	0.222	0.328	0.131	0.262
Weather	**0.030**	**0.045**	**0.030**	0.054	0.052	0.110	0.076	0.171	0.099	0.203	0.032	0.059	0.031	0.057	0.045	0.104	0.055	0.117

Table 3
For the long-term forecasting task, the historical length is set as 36 for ILI and 96 for the others. The experiments were repeated three times and we reported the mean values. All the results are averaged from 4 different prediction lengths, that is {24, 36, 48, 60} for ILI and {96, 192, 336, 720} for the others. Full results are provided in Appendix A.2.

Model	Lena-TRNN		TimesNet		Dlinear		ETSformer		FEDformer		Stationary		Autoformer		Informer		LightTS	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	0.420	0.410	**0.400**	**0.406**	0.403	0.407	0.429	0.425	0.448	0.452	0.481	0.456	0.588	0.517	0.961	0.734	0.435	0.437
ETTm2	**0.291**	**0.327**	**0.291**	0.333	0.350	0.401	0.293	0.342	0.305	0.349	0.306	0.347	0.327	0.371	1.410	0.810	0.409	0.436
ETTh1	0.463	**0.443**	0.458	0.450	0.456	0.452	0.542	0.510	**0.440**	0.460	0.570	0.537	0.496	0.487	1.040	0.795	0.491	0.479
ETTh2	**0.380**	**0.398**	0.414	0.427	0.559	0.515	0.439	0.452	0.437	0.449	0.526	0.516	0.450	0.459	4.431	1.729	0.602	0.543
ECL	**0.169**	**0.263**	0.192	0.295	0.212	0.300	0.208	0.323	0.214	0.327	0.193	0.296	0.227	0.338	0.311	0.397	0.229	0.329
Traffic	**0.560**	0.337	0.620	**0.336**	0.625	0.383	0.621	0.396	0.610	0.376	0.624	0.340	0.628	0.379	0.764	0.416	0.622	0.392
Weather	0.261	**0.284**	**0.259**	0.287	0.265	0.317	0.271	0.334	0.309	0.360	0.288	0.314	0.338	0.382	0.634	0.548	0.261	0.312
Exchange	0.376	0.415	0.416	0.443	**0.354**	**0.414**	0.410	0.427	0.519	0.500	0.461	0.454	0.613	0.539	1.550	0.998	0.385	0.447
ILI	2.147	**0.889**	2.139	0.931	2.616	1.090	2.497	1.004	2.847	1.144	**2.077**	0.914	3.006	1.161	5.137	1.544	7.382	2.003

Table 4
Compared results with probabilistic forecasting methods on $CRPS_{sum}$ (lower is better). The baseline results are cited from the original paper.

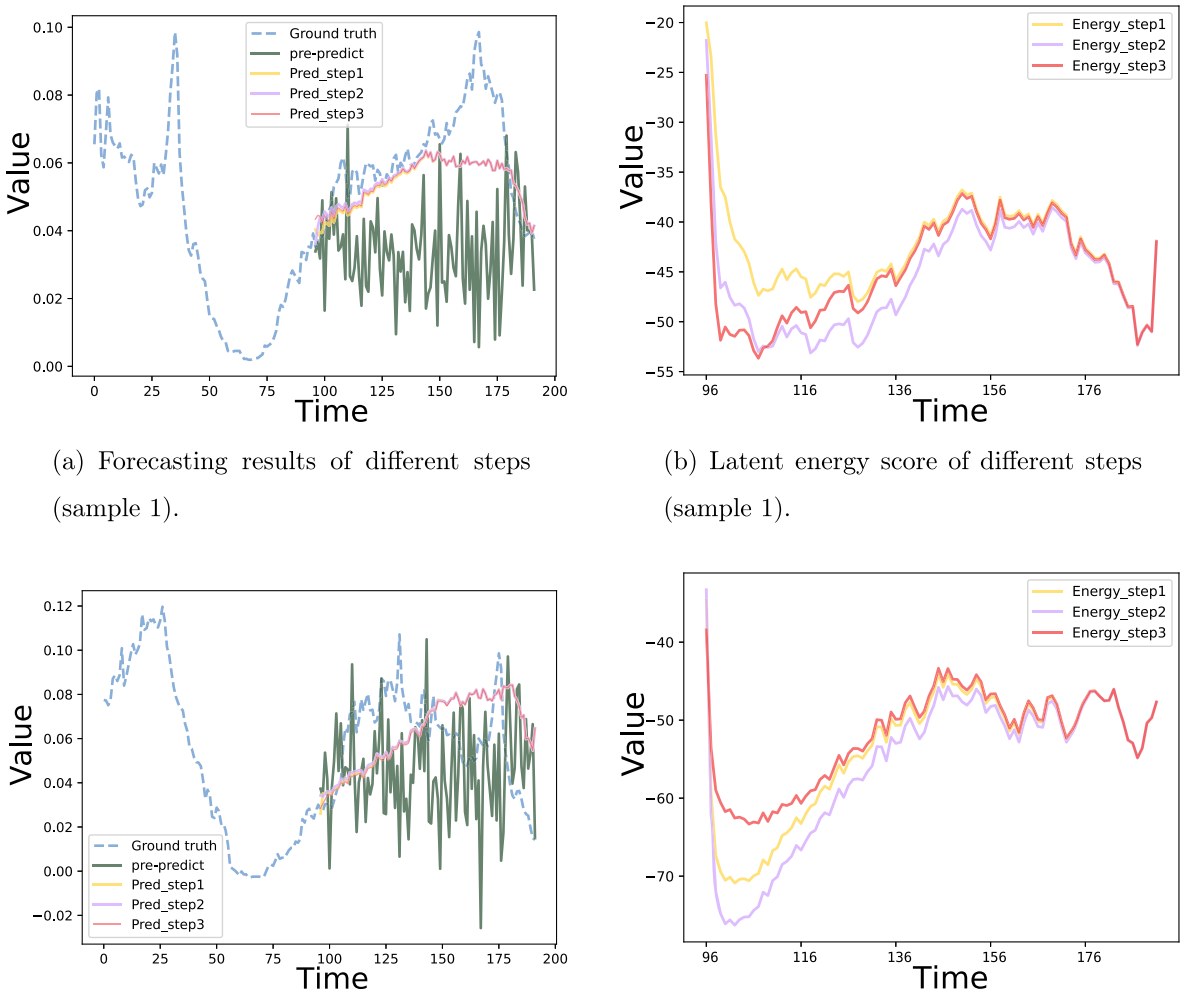
	Solar	Exchange	Traffic
Transformer-MAF	0.301	0.005	0.056
TimeGrad	0.287	0.006	0.049
ScoreGrad	**0.256**	0.006	0.037
Lena-TRNN	0.261	**0.002**	**0.010**

evaluation, we compute the sum of Continuous Ranked Probability Score (CRPS) for comparison, which is the distribution of the sum of all time series across K features. The quantitative results in Table 4 show that our method also achieves competitive performance compared with the other three probabilistic forecasting models, though Lena-TRNN is designed for deterministic forecasting.

4.5. Empirical analysis of energy flow

We further visualize the predicted results and the explicitly learned energy score to facilitate a better understanding and demonstration of the guiding capability of energy.

We randomly select two complete blocks of missing data with a mask ratio of 25 % to illustrate the imputed results and their corresponding energy flow. As shown in Fig. 3, the updating step of our energy optimization is 3. The blue dashed line represents the missing part of the original series and the 'partial_input' represents the partially observed input series with zeros padding. The partially observed input series is out-of-distribution, reflected by the high energy score by the green line in Fig. 3. Through our energy optimization, the imputed results are good



(a) Forecasting results of different steps (sample 1).

(b) Latent energy score of different steps (sample 1).

(c) Forecasting results of different steps (sample 2).

(d) Latent energy scores of different steps (sample 2).

Fig. 4. Visualization of long-term forecasting results with input-96-predict-96 and corresponding latent energy scores of different updating steps on the Weather dataset.

enough with one-step updating shown in Fig. 3. As the energy score decreases, the predicted results are refined along the direction of energy minimization, and 'Pred_step3' gets closer to the ground truth. This also demonstrates that by iteratively optimizing the gradient along the direction of energy minimization, we can easily obtain predicted samples and explicitly learn the inherent energy flow.

Moreover, we take the input-96-predict-96 on the Weather dataset to demonstrate the guiding capability of energy flow in long-term forecasting. Fig. 4 shows two samples of forecasting results and corresponding energy scores of the target series. We set the updating step N as 2 to train our Lena-TRNN. 'pre-predict' represents the initial prediction without energy updating and 'Pred_step1,2,3' represents the predicted results after updating step1, step2, and step3. 'Energy_step1,2,3' represents the energy score with the corresponding updating step. From Fig. 4(b) and (d), we can see that the energy score of 'step2' is the lowest, which is consistent with the predicted results in Fig. 4(a) and (c). When increasing the updating step, the energy score gets higher and the predicted result does not get better. This phenomenon indicates that our method can get excellent results with two-step energy flow optimization. Furthermore, the energy score is lower in the beginning and second half of the forecasting range, indicating that the predicted results are closer to the ground truth seen from Figs. 4(a) and 4(b). In another case, as seen from Fig. 4(c) and (d), the predictive results are closer to the ground truth with a lower energy score in the initial range of 96–110. However, the energy score gets higher as the prediction time increases, leading to invalid predicted results. It is worth mentioning that despite the predicted curve having several intersection points with the ground truth, the consistently high energy scores throughout the entire erroneous prediction trend indicate that the energy flow plays an indicative role in the effectiveness of the predicted outcomes.

4.6. Ablation study

We further conduct the ablation study to verify the effectiveness of our proposed energy flow modeling for long-term forecasting and imputation tasks on the ECL, Weather, and ETTh1 datasets. Table 5 shows the quantitative results averaged on the four prediction lengths and Table 6 shows the quantitative results averaged on the four missing rates. We can see that our full model achieves the best performance on both the long-term forecasting and imputation tasks. Without any designed component, a decline can be observed in forecasting performance on all three datasets. These results indicate that focusing on the inherent energy flow of time series data could help to better capture the intricate temporal patterns and improve forecasting

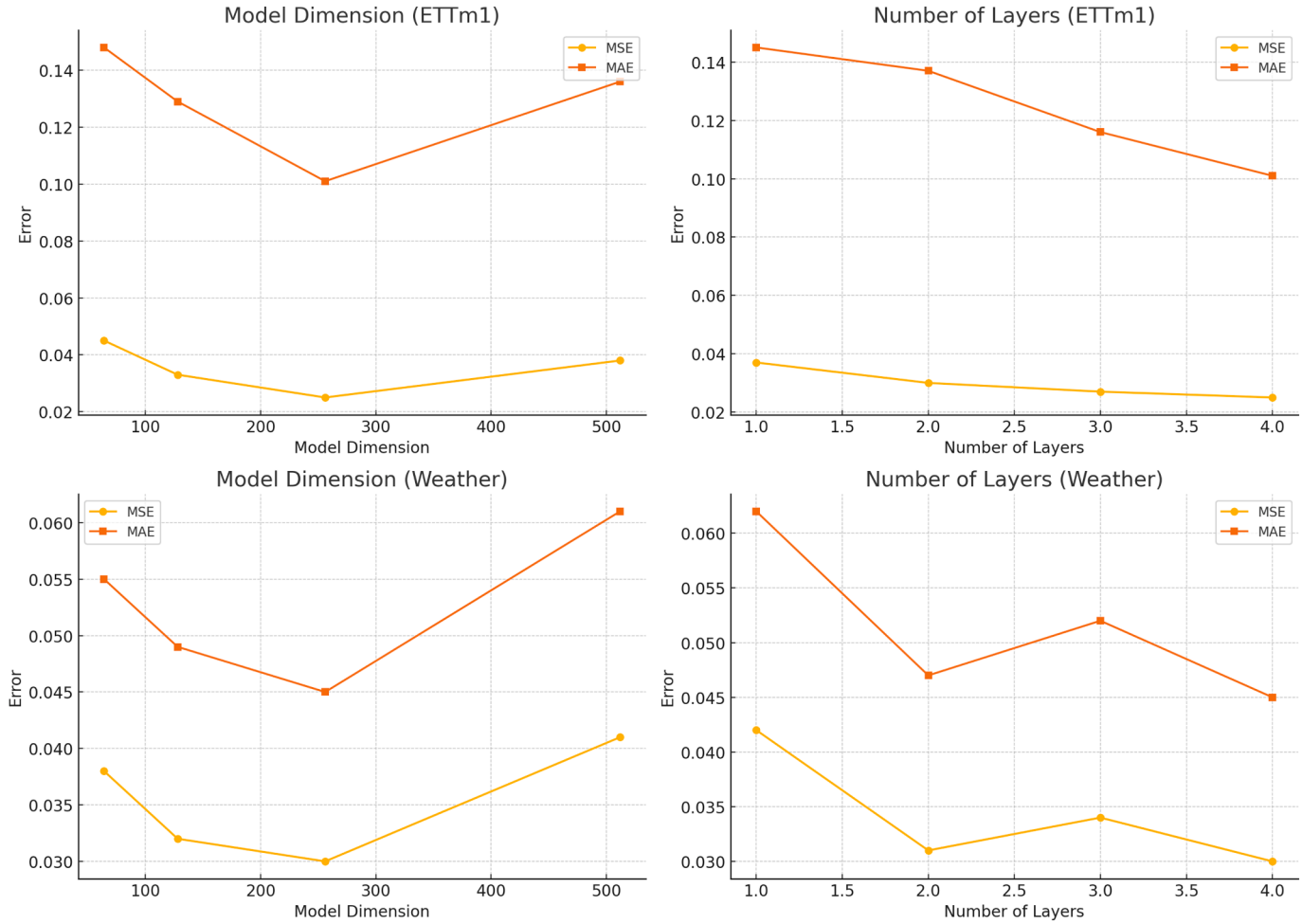


Fig. 5. Sensitivity analysis of model dimension and number of layers of our Lena-TRNN.

Table 5
Ablation study for long-term forecasting on ECL, Weather, and ETTh1 datasets. We eliminated our designed components accordingly.

Component			Dataset					
Energy estimation	Transformer block	GRU net	ECL		Weather		ETTh1	
			MSE	MAE	MSE	MAE	MSE	MAE
✓	✓	✓	**0.169**	**0.263**	**0.261**	**0.284**	**0.463**	**0.443**
-	✓	✓	0.203	0.296	0.309	0.313	0.503	0.470
-	-	✓	0.238	0.319	0.280	0.297	0.493	0.464
-	-	-	0.254	0.337	0.292	0.311	0.482	0.461

Table 6
Ablation study for imputation on ECL, Weather, and ETTh1 datasets. We eliminated our designed components accordingly.

Component			Dataset					
Energy estimation	Transformer block	GRU net	ECL		Weather		ETTh1	
			MSE	MAE	MSE	MAE	MSE	MAE
✓	✓	✓	**0.094**	**0.211**	**0.029**	**0.046**	**0.074**	**0.182**
–	✓	✓	0.131	0.255	0.039	0.062	0.167	0.227
–	–	✓	0.134	0.258	0.042	0.067	0.191	0.254
–	–	–	0.108	0.229	0.035	0.058	0.124	0.220

accuracy in industrial applications. In addition, the Transformer block takes effect in capturing the global dependence of the whole time series data.

4.7. Sensitivity analysis

We conducted a sensitivity analysis of the model dimension and the number of layers of our Lena-TRNN on ETTm1 and Weather datasets as examples for the imputation task. As shown in Fig. 5, our method is a little sensitive to the model dimension and achieves the best results when the model dimension is 256. Regarding the number of layers, we selected 4 layers in our settings for improved performance.

5. Conclusion

This paper focuses on exploring the inherent energy flow of time series data and presents a novel energy-based framework for multivariate time series forecasting and imputation. This brand-new perspective allows us to discover the relation between energy flow and temporal variation and better recognize the underlying distribution of the time-series data. We explicitly learn the trends of energy flow and guide the model's prediction with energy optimization. Our method achieves new state-of-the-art results on extensive experiments and shows the superiority of exploring the inherent energy flow of time series data. In the future, we will explore the connection between energy variation and adversarial examples to enhance model robustness and generalization.

CRediT authorship contribution statement

Penglei Gao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation; **Rui Zhang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition; **Xi Yang:** Writing – review & editing, Validation, Investigation; **Zhuang Qian:** Writing – review & editing, Software, Data curation; **Kaizhu Huang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Full results

We provide the full quantitative results of imputation and long-term forecasting tasks in the Appendix Section in Table A.1 and A.2. Our method achieves the best results on most tasks.

Table A.1
Full results of the imputation task. The data are randomly masked with ratios of {12.5 %, 25 %, 37.5 %, 50 %} in length-96 time series. The best results are marked underline and the second-best results are marked bold.

Model		L ena-TRNN (Ours)		TimesNet		DLinear		ETSformer		FEDformer		Stationary		Autoformer		Informer		LightTS	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	12.5 %	0.017	0.085	**0.019**	**0.092**	0.058	0.162	0.067	0.188	0.035	0.135	0.026	0.107	0.034	0.124	0.047	0.155	0.075	0.180
	25 %	0.021	0.095	**0.023**	**0.101**	0.080	0.193	0.096	0.229	0.052	0.166	0.032	0.119	0.046	0.144	0.063	0.18	0.093	0.206
	37.5 %	0.028	0.108	**0.029**	**0.111**	0.103	0.219	0.133	0.271	0.069	0.191	0.039	0.131	0.057	0.161	0.079	0.200	0.113	0.231
	50 %	0.034	0.118	**0.036**	**0.124**	0.132	0.248	0.186	0.323	0.089	0.218	0.047	0.145	0.067	0.174	0.093	0.218	0.134	0.255
	Avg	0.025	0.101	**0.027**	**0.107**	0.093	0.206	0.120	0.253	0.062	0.177	0.036	0.126	0.051	0.15	0.071	0.188	0.104	0.218
ETTm2	12.5 %	0.017	0.072	**0.018**	**0.080**	0.062	0.166	0.108	0.239	0.056	0.159	0.021	0.088	0.023	0.092	0.133	0.270	0.034	0.127
	25 %	0.019	0.077	**0.020**	**0.085**	0.085	0.196	0.164	0.294	0.080	0.195	0.024	0.096	0.026	0.101	0.135	0.272	0.042	0.143
	37.5 %	0.022	0.086	**0.023**	**0.091**	0.106	0.222	0.237	0.356	0.110	0.231	0.027	0.103	0.030	0.108	0.155	0.293	0.051	0.159
	50 %	0.025	0.094	**0.026**	**0.098**	0.131	0.247	0.323	0.421	0.156	0.276	0.030	0.108	0.035	0.119	0.200	0.333	0.059	0.174
	Avg	0.021	0.082	**0.022**	**0.088**	0.096	0.208	0.208	0.327	0.101	0.215	0.026	0.099	0.029	0.105	0.156	0.292	0.046	0.151
ETTh1	12.5 %	0.047	0.147	**0.057**	**0.159**	0.151	0.267	0.126	0.263	0.070	0.190	0.060	0.165	0.074	0.182	0.114	0.234	0.240	0.345
	25 %	0.065	0.173	**0.069**	**0.178**	0.180	0.292	0.169	0.304	0.106	0.236	0.080	0.189	0.090	0.203	0.140	0.262	0.265	0.364
	37.5 %	0.082	0.193	**0.084**	**0.196**	0.215	0.318	0.220	0.347	0.124	0.258	0.102	0.212	0.109	0.222	0.174	0.293	0.296	0.382
	50 %	0.102	**0.216**	0.102	0.215	0.257	0.347	0.293	0.402	0.165	0.299	0.133	0.240	0.137	0.248	0.215	0.325	0.334	0.404
	Avg	0.074	0.182	**0.078**	**0.187**	0.201	0.306	0.202	0.329	0.117	0.246	0.094	0.201	0.103	0.214	0.161	0.279	0.284	0.373
ETTh2	12.5 %	0.037	0.120	**0.040**	**0.130**	0.100	0.216	0.187	0.319	0.095	0.212	0.042	0.133	0.044	0.138	0.305	0.431	0.101	0.231
	25 %	0.043	0.131	**0.046**	**0.141**	0.127	0.247	0.279	0.390	0.137	0.258	0.049	0.147	0.050	0.149	0.322	0.444	0.115	0.246
	37.5 %	0.048	0.140	**0.052**	**0.151**	0.158	0.276	0.400	0.465	0.187	0.304	0.056	0.158	0.060	0.163	0.353	0.462	0.126	0.257
	50 %	0.055	0.153	**0.060**	**0.162**	0.183	0.299	0.602	0.572	0.232	0.341	0.065	0.170	0.068	0.173	0.369	0.472	0.136	0.268
	Avg	0.046	0.136	**0.049**	**0.146**	0.142	0.259	0.367	0.436	0.163	0.279	0.053	0.152	0.055	0.156	0.337	0.452	0.119	0.250
ECL	12.5 %	**0.087**	**0.203**	0.085	0.202	0.092	0.214	0.196	0.321	0.107	0.237	0.093	0.210	0.089	0.210	0.218	0.326	0.102	0.229
	25 %	**0.091**	**0.208**	0.089	0.206	0.118	0.247	0.207	0.332	0.120	0.251	0.097	0.214	0.096	0.220	0.219	0.326	0.121	0.252
	37.5 %	**0.096**	0.213	0.094	0.213	0.144	0.276	0.219	0.344	0.136	0.266	0.102	0.220	0.104	0.229	0.222	0.328	0.141	0.273
	50 %	**0.102**	0.221	0.100	0.221	0.175	0.305	0.235	0.357	0.158	0.284	0.108	0.228	0.113	0.239	0.228	0.331	0.160	0.293
	Avg	**0.094**	**0.211**	0.092	0.210	0.132	0.26	0.214	0.339	0.130	0.259	0.100	0.218	0.101	0.225	0.222	0.328	0.131	0.262
Weather	12.5 %	**0.026**	0.040	0.025	**0.045**	0.039	0.084	0.057	0.141	0.041	0.107	0.027	0.051	0.026	0.047	0.037	0.093	0.047	0.101
	25 %	0.028	0.044	**0.029**	**0.052**	0.048	0.103	0.065	0.155	0.064	0.163	0.029	0.056	0.030	0.054	0.042	0.100	0.052	0.111
	37.5 %	0.030	0.046	**0.031**	**0.057**	0.057	0.117	0.081	0.180	0.107	0.229	0.033	0.062	0.032	0.060	0.049	0.111	0.058	0.121
	50 %	0.034	0.051	0.034	**0.062**	0.066	0.134	0.102	0.207	0.183	0.312	0.037	0.068	0.037	0.067	0.053	0.114	0.065	0.133
	Avg	0.030	0.045	0.030	**0.054**	0.052	0.110	0.076	0.171	0.099	0.203	0.032	0.059	0.031	0.057	0.045	0.104	0.055	0.117
Count		50/**10**		15/**45**		0/**0**		0/**0**		0/**0**		0/**0**		0/**0**		0/**0**		0/**0**	

Table A.2
Full results of the long-term forecasting task. The historical length is set as 36 for ILI and 96 for the others. We conduct the experiments with four different prediction lengths, that is {24, 36, 48, 60} for ILI and {96, 192, 336, 720} for the others. The best results are marked underline and the second-best results are marked bold.

Model		L ena-TRNN (Ours)		TimesNet		Dlinear		ETSformer		FEDformer		Stationary		Autoformer		Informer		LightTS	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	0.375	0.394	0.338	**0.375**	**0.345**	0.372	0.375	0.398	0.379	0.419	0.386	0.398	0.505	0.475	0.672	0.571	0.374	0.400
	192	0.384	0.386	0.374	**0.387**	**0.380**	0.389	0.408	0.410	0.426	0.441	0.459	0.444	0.553	0.496	0.795	0.669	0.400	0.407
	336	0.426	0.411	0.410	0.411	**0.413**	0.413	0.435	0.428	0.445	0.459	0.495	0.464	0.621	0.537	1.212	0.871	0.438	0.438
	720	0.497	0.449	**0.478**	**0.450**	0.474	0.453	0.499	0.462	0.543	0.490	0.585	0.516	0.671	0.561	1.166	0.823	0.527	0.502
	avg	0.420	0.410	0.400	0.406	**0.403**	**0.407**	0.429	0.425	0.448	0.452	0.481	0.456	0.588	0.517	0.961	0.734	0.435	0.437
ETTm2	96	0.183	0.259	**0.187**	**0.267**	0.193	0.292	0.189	0.280	0.203	0.287	0.192	0.274	0.255	0.339	0.365	0.453	0.209	0.308
	192	**0.252**	0.304	0.249	**0.309**	0.284	0.362	0.253	0.319	0.269	0.328	0.280	0.339	0.281	0.340	0.533	0.563	0.311	0.382
	336	0.312	0.342	**0.321**	**0.351**	0.369	0.427	0.314	0.357	0.325	0.366	0.334	0.361	0.339	0.372	1.363	0.887	0.442	0.466
	720	0.419	0.403	0.408	0.403	0.554	0.522	**0.414**	0.413	0.421	0.415	0.417	0.413	0.433	0.432	3.379	1.338	0.675	0.587
	avg	0.291	0.327	**0.291**	**0.333**	0.350	0.401	0.293	0.342	0.305	0.349	0.306	0.347	0.327	0.371	1.410	0.810	0.409	0.436
ETTh1	96	0.407	0.406	**0.384**	**0.402**	0.386	0.400	0.494	0.479	0.376	0.419	0.513	0.491	0.449	0.459	0.865	0.713	0.424	0.432
	192	0.462	0.438	**0.436**	0.429	0.437	**0.432**	0.538	0.504	0.420	0.448	0.534	0.504	0.500	0.482	1.008	0.792	0.475	0.462
	336	0.493	0.456	0.491	0.469	**0.481**	**0.459**	0.574	0.521	0.459	0.465	0.588	0.535	0.521	0.496	1.107	0.809	0.518	0.488
	720	0.489	0.471	0.521	**0.500**	0.519	0.516	0.562	0.535	**0.506**	0.507	0.643	0.616	0.514	0.512	1.181	0.865	0.547	0.533
	avg	0.463	0.443	0.458	**0.450**	**0.456**	0.452	0.542	0.510	0.440	0.460	0.570	0.537	0.496	0.487	1.040	0.795	0.491	0.479
ETTh2	96	0.293	0.338	0.340	**0.374**	**0.333**	0.387	0.340	0.391	0.358	0.397	0.476	0.458	0.346	0.388	3.755	1.525	0.397	0.437
	192	0.381	0.392	**0.402**	**0.414**	0.477	0.476	0.430	0.439	0.429	0.439	0.512	0.493	0.456	0.452	5.602	1.931	0.520	0.504
	336	0.421	0.425	**0.452**	**0.452**	0.594	0.541	0.485	0.479	0.496	0.487	0.552	0.551	0.482	0.486	4.721	1.835	0.626	0.559
	720	0.423	0.439	**0.462**	**0.468**	0.831	0.657	0.500	0.497	0.463	0.474	0.562	0.560	0.515	0.511	3.647	1.625	0.863	0.672
	avg	0.380	0.398	**0.414**	**0.427**	0.559	0.515	0.439	0.452	0.437	0.449	0.526	0.516	0.450	0.459	4.431	1.729	0.602	0.543
ECL	96	0.145	0.240	**0.168**	**0.272**	0.197	0.282	0.187	0.304	0.193	0.308	0.169	0.273	0.201	0.317	0.274	0.368	0.207	0.307
	192	0.157	0.250	0.184	0.289	0.196	**0.285**	0.199	0.315	0.201	0.315	**0.182**	0.286	0.222	0.334	0.296	0.386	0.213	0.316
	336	0.171	0.266	**0.198**	**0.300**	0.209	0.301	0.212	0.329	0.214	0.329	0.200	0.304	0.231	0.338	0.300	0.394	0.230	0.333
	720	0.205	0.296	**0.220**	**0.320**	0.245	0.333	0.233	0.345	0.246	0.355	0.222	0.321	0.254	0.361	0.373	0.439	0.265	0.360
	avg	0.169	0.263	**0.192**	**0.295**	0.212	0.300	0.208	0.323	0.214	0.327	0.193	0.296	0.227	0.338	0.311	0.397	0.229	0.329
Traffic	96	0.601	0.355	**0.593**	0.321	0.650	0.396	0.607	0.392	0.587	0.366	0.612	**0.338**	0.613	0.388	0.719	0.391	0.615	0.391
	192	0.593	0.356	0.617	0.336	**0.598**	0.370	0.621	0.399	0.604	0.373	0.613	**0.340**	0.616	0.382	0.696	0.379	0.601	0.382
	336	0.513	0.313	0.629	0.336	**0.605**	0.373	0.622	0.396	0.621	0.383	0.618	**0.328**	0.622	0.337	0.777	0.420	0.613	0.386
	720	0.532	0.324	0.640	**0.350**	0.645	0.394	0.632	0.396	**0.626**	0.382	0.653	0.355	0.660	0.408	0.864	0.472	0.658	0.407
	avg	0.560	**0.337**	0.620	0.336	0.625	0.383	0.621	0.396	**0.610**	0.376	0.624	0.340	0.628	0.379	0.764	0.416	0.622	0.392
Weather	96	0.170	0.212	**0.172**	**0.220**	0.196	0.255	0.197	0.281	0.217	0.296	0.173	0.223	0.266	0.336	0.300	0.384	0.182	0.242
	192	**0.226**	**0.263**	0.219	0.261	0.237	0.296	0.237	0.312	0.276	0.336	0.245	0.285	0.307	0.367	0.598	0.544	0.227	0.287
	336	**0.282**	0.304	0.280	**0.306**	0.283	0.335	0.298	0.353	0.339	0.380	0.321	0.338	0.359	0.395	0.578	0.523	0.282	0.334
	720	0.365	0.357	0.365	**0.359**	0.345	0.381	**0.352**	0.388	0.403	0.428	0.414	0.410	0.419	0.428	1.059	0.741	**0.352**	0.386
	avg	**0.261**	0.284	0.259	**0.287**	0.265	0.317	0.271	0.334	0.309	0.360	0.288	0.314	0.338	0.382	0.634	0.548	**0.261**	0.312
Exchange	96	0.092	**0.214**	0.107	0.234	**0.088**	0.218	0.085	0.204	0.148	0.278	0.111	0.237	0.197	0.323	0.847	0.752	0.116	0.262
	192	**0.182**	**0.305**	0.226	0.344	0.176	0.315	**0.182**	0.303	0.271	0.380	0.219	0.335	0.300	0.369	1.204	0.895	0.215	0.359
	336	0.351	0.431	0.367	0.448	0.313	0.427	**0.348**	**0.428**	0.460	0.500	0.421	0.476	0.509	0.524	1.672	1.036	0.377	0.466
	720	0.880	0.710	0.964	0.746	**0.839**	0.695	1.025	0.774	1.195	0.841	1.092	0.769	1.447	0.941	2.478	1.310	0.831	**0.699**
	avg	**0.376**	**0.415**	0.416	0.443	0.354	0.414	0.410	0.427	0.519	0.500	0.461	0.454	0.613	0.539	1.550	0.998	0.385	0.447
ILI	96	2.364	0.930	**2.317**	**0.934**	2.398	1.040	2.527	1.020	3.228	1.260	2.294	0.945	3.483	1.287	5.764	1.677	8.313	2.144
	192	2.063	**0.868**	**1.972**	0.920	2.646	1.088	2.615	1.007	2.679	1.080	1.825	0.848	3.103	1.148	4.755	1.467	6.631	1.902
	336	**2.109**	0.872	2.238	0.940	2.614	1.086	2.359	0.972	2.622	1.078	2.010	**0.900**	2.669	1.085	4.763	1.469	7.299	1.982
	720	**2.053**	0.885	2.027	**0.928**	2.804	1.146	2.487	1.016	2.857	1.157	2.178	0.963	2.770	1.125	5.264	1.564	7.283	1.985
	avg	2.147	0.889	**2.139**	0.931	2.616	1.090	2.497	1.004	2.847	1.144	2.077	**0.914**	3.006	1.161	5.137	1.544	7.382	2.003
Count		50/**14**		18/**45**		10/**15**		3/**5**		5/**3**		5/**6**		0/**0**		0/**0**		1/**3**	

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